Numerical Simulation of Tunnel Gas Explosion and Structural Damage Analysis Based on ANSYS/LS-DYNA

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Abstract

To investigate the damage and destruction caused by gas explosion shock waves to the wall structures of a roadway, the physical and mathematical models of gas explosion in roadways were established using ANSYS/LS-DYNA to conduct a numerical simulation study on the destructive characteristics of gas explosion shock waves in excavation roadways. The results indicate that the Excess pressure measurements are higher at the edge and center positions of the roadway wall, resulting in relatively more severe damage to the wall. The shock waves also exhibit reflection and superposition on the axial wall of the roadway, leading to fluctuations in the overall Excess pressure peak. After the gas explosion, the shock waves propagate towards the opening direction, with the maximum load applied to the wall in the gas zone, gradually spreading to the air zone. As the energy of the explosion shock waves decays, the stress continues to be applied to the wall structure, with concentrated pressure exerting static damage to the wall structure. Ultimately, the damage exceeds the structure's bearing capacity, resulting in the instability and destruction of the roadway. These research findings provide theoretical references for optimizing the design of roadway structures.

CCS Concepts

• **General and reference** → Document types; General conference proceedings.

Keywords

Gas explosion, Excess pressure, Shock wave, Numerical simulation

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1 Introduction

Gas explosion accidents are one of the severe disasters in coal mine production, causing significant property losses and casualties. According to statistics, major gas explosion accidents occurring in roadways account for approximately 30.5% of the total number of major coal mine accidents in China since the founding of the country, and significant gas explosion accidents account for approximately 27.1% of the total number of major coal mine accidents in China since 2000. Roadways are high-risk locations for major gas explosion accidents in underground coal mines[1]. Studying the propagation law and damage pattern during gas explosion in roadways will help formulate corresponding explosion-proof and suppression measures to reduce the damage caused by gas explosions to mine roadways.

Currently, numerous scholars have conducted extensive theoretical analysis, numerical simulations, and experimental studies on the propagation laws of gas explosions in tunnels. Most research focuses on explosion pressure and flame propagation patterns[2-5], with few studies on the damage and failure processes of wall structures under gas explosions. To further investigate the damage process of gas explosions in tunnels, physical and mathematical models of tunnel gas explosions have been established. Using the nonlinear dynamic analysis method of ANSYS/LS-DYNA software, a typical tunnel structure model was taken as the research object. Through analyzing the failure processes of closed end faces and axial wall structures, a more intuitive understanding of the wall damage process was obtained, displaying the changes in explosion Excess pressure and wall cloud diagrams at different times. The numerical simulation study on the propagation laws and failure characteristics of gas explosion shock waves in tunnels reveals the patterns of gas explosion damage to tunnel walls.Thereby, theoretical basis can be provided for reinforcement design of tunnel structures and formulation of anti-explosion and explosion suppression measures.

2 Establishment of gas explosion calculation model

2.1 Mathematical model

2.1.1 Basic assumptions. Gas explosion is an extremely complex physical and chemical reaction involving a large number of instantaneous and intermediate products. To simplify the calculation, the intermediate products of the explosion process are ignored. The gas explosion process is an irreversible reaction, and the specific heat capacity of the gas mixture in the tunnel follows the mixing rule. Only the heat source of gas explosion exists in the tunneling tunnel, and the heat exchange between the tunnel and the outside is not

considered. The tunnel model wall is regarded as a rigid wall, and the tunnel space is under standard atmospheric pressure conditions. The initial state temperature and pressure of the gas are uniformly distributed.

2.1.2 Basic Control Equations. In the gas explosion of tunneling roadway, it is the flow process of explosion shock wave in a confined space with one end closed and the other end open. In the rectangular coordinate system, it satisfies the law of mass conservation, the law of energy conservation, the law of momentum conservation, and the state equation. The conservation equations for three-dimensional flow during the gas explosion process are as follows.

Mass conservation equation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} + \frac{\partial (\rho v)}{\partial y} + \frac{\partial (\rho w)}{\partial z} = 0 \tag{1}$$

Momentum Conservation Equation:

$$\begin{cases} \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} = -\frac{1}{\rho}\frac{\partial \rho}{\partial x} \\ \frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} = -\frac{1}{\rho}\frac{\partial \rho}{\partial y} \\ \frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} = -\frac{1}{\rho}\frac{\partial \rho}{\partial z} \end{cases} \tag{2}$$

The energy conservation equation:

$$\frac{\partial e}{\partial t} + u\frac{\partial e}{\partial x} + v\frac{\partial e}{\partial y} + w\frac{\partial e}{\partial z} = 0 \tag{3}$$

State equation:

$$p = p(\rho, T) = \rho RT \tag{4}$$

In the equation: x, y, z are coordinate parameters; ρ represents fluid density, kg/m^3; p is pressure, Pa; u, v, w are the velocities in the three coordinate directions, respectively, m/s; t is the time coordinate; T represents temperature, K; R is the gas constant; e is specific energy, kJ/kg, $e = \frac{p}{\gamma - 1} + \frac{\rho(u^2+v^2+w^2)}{2}$, where γ is the gas index, which is a constant.

2.2 Physical Model

2.2.1 Establishment of Finite Element Model. The ALE modeling method in ANSYS/LS-DYNA can realize the simulation analysis of the damage and destruction caused by explosive shock waves to the roadway walls. Therefore, 3D SOLID 164 solid element modeling is adopted for gas, air, and roadway, and the explicit dynamic element type included in ANSYS/LS-DYNA is a linear element, which can be well applied to problems involving large deformations and material failures. The physical model is shown in Figure 1.

Based on the symmetry of the model, a 1/2 excavation roadway gas explosion model is established for analysis, and a fluid-solid coupling numerical model of "gas-air-roadway wall structure" is constructed. The finite element model is shown in Figure 2. The location of the explosion source is at the front end of the roadway, and the model dimensions are 10 m × 2 m × 1 m (length × width × height). The thickness of the roadway wall is 0.2 m, and the excavation roadway has an open structure at one end and a closed structure at the other end. The premixed gas of gas and air is separated by a membrane, and the boundary condition of the roadway is a non-reflective boundary to simulate an infinite domain.

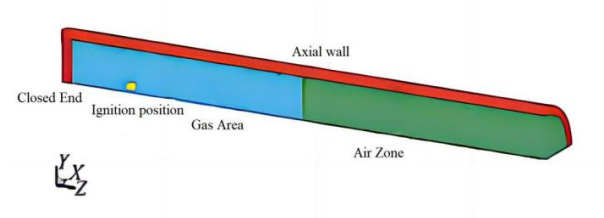

Figure 1: Roadway physical model.

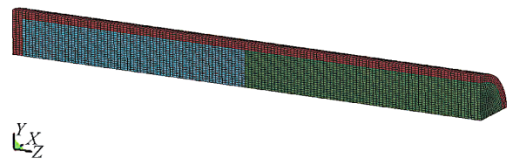

Figure 2: Finite element model.

2.2.2 Selection of Material Parameters and State Equation. The air material model employs *MAT_NULL, and its equation of state is the linear polynomial *EOS_LINEAR_POLYNOMIAL, representing the linear relationship of internal energy within the unit initial volume. The equation of state for the linear polynomial is:

$$P = C_0 + C_1\mu + C_2\mu^2 + C_3\mu^3 + \left(C_4 + C_5\mu + C_6\mu^2\right)\rho_0 E \tag{5}$$

In Equation 5): $\mu = \rho/\rho_0 - 1$; C_0 to C_6 are the parameters of the state equation, which are defined constants; ρ is the current density, kg/m^3; ρ_0 is the initial density, kg/m^3; ρ/ρ_0 is the relative density; E is the internal energy of the material. The calculation parameters for air are shown in Table 1.

In the simulation, the explosion model adopts *MAT_HIGH_EXPLOSIVE_BURNJWL, and the equation of state is *EOS_JWL (Jones-Wilkins-Lee). The p-V relationship of the equation of state is as follows:

$$p = A\left(1 - \frac{\omega}{A_1 v}\right)e^{-R_1 V} + B\left(1 - \frac{\omega}{R_2 V}\right)e^{-R_2 V} + \frac{\omega E_0}{V} \tag{6}$$

In Equation 6): p represents the detonation pressure in Pa; V represents the relative volume in m^3; E_0 represents the initial internal energy density in kJ/m^3; and parameters A, B, R_1, R_2, and ω are material constants determined by detonation experiments. Under the impact of the gas explosion shock wave, the tunnel wall will experience wall damage evolution and crack propagation, generating tensile stress on the surface, leading to tensile failure of the wall. Therefore, the H-J-C (Holmquist-Johnson-Cook) model is adopted for the wall structure, and the tunnel wall parameters are shown in Table 2 [6-8].

3 Numerical Simulation and Result Analysis

3.1 Wall Excess pressure Analysis

3.1.1 Analysis of Excess pressure on the Closed-end Wall. A series of target points were selected on the closed-end wall of the excavated tunnel to record the parameter changes during the destruction process of the explosion shock wave, as shown in Figure 3. Along

Table 1: Air calculation parameters.

Parameters	$\rho_0/(kg\cdot m^{-3})$	C_0/Pa	C_1	C_2	C_3	C_4	C_5	C_6	$E/(J\cdot m^{-3})$
Value Extraction	1.29	-1×10^5	0	0	0	0.4	0.4	0	2.5×10^5

Table 2: Roadway wall parameters.

$\rho_c/(kg\cdot m^{-3})$	E_c/GPa	V_c	G_c/GPa	K	m	$K_{IC}/(MPa^*m^{1/2})$	σ_{cy}/MPa	f_s	f_d
2.27	12.60	0.22	5.16	$5.75\cdot10^{-11}$	6.0	2.07	35	1.5	0.3

Note: ρ_c represents the density of the tunnel wall; E_c is the elastic modulus; V_c is the Poisson's ratio; G_c is the shear modulus; K, m, and K_{IC} are parameters for wall performance; σ_{cy} stands for compressive strength; f_s indicates the failure strain of the wall; and f_d represents the damage degree value.

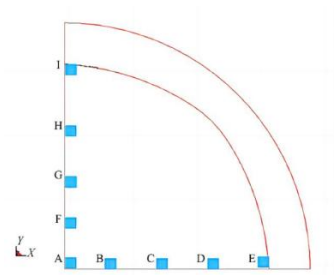

Figure 3: Excess pressure data acquisition point at closed-end.

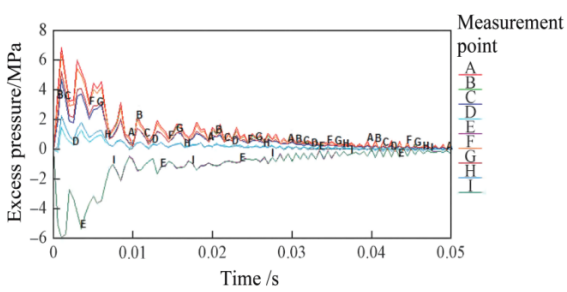

Figure 4: Excess pressure time curve of closed-end wall.

the height direction (X-axis) within the tunnel, a collection point was chosen every 0.2 meters, totaling 5 collection points. Along the lateral length direction of the tunnel (Y-axis), a data collection point was selected every 0.2 meters, with a total of 9 data collection points. The Excess pressure time history curves of each measurement point at different time instants are shown in Figure 4.

As shown in Figure 3, the change trends of measurement points A, B, C, D, F, G, and H are consistent. Additionally, the Excess pressure change trends of measurement points E and I are largely similar. The overall change in the peak Excess pressure in the enclosed roadway initially increases and then decreases sharply. After reflecting from the wall surface, it reaches the peak again, and after repeated reflections, it tends to stabilize at normal pressure. Since the explosion shock wave does not propagate in a stable manner but constantly changes during propagation, the damage to the roadway in different directions may vary. From the perspective of the maximum Excess pressure peak values at the measurement points at the closed end, measurement point A has the highest value, followed by measurement points B, E, and C, and finally measurement point D. Among them, the maximum value at measurement point A is 6.92 MPa, and the Excess pressure peak value at measurement point E is 5.98 MPa. This is due to the intersection of measurement point E with the two wall surfaces, resulting in an increased Excess pressure peak. Analysis shows that the Excess pressure measurements at the edge and center positions of the roadway wall surface are relatively higher, indicating more severe wall damage in these locations.

Therefore, due to the mutual constraints of the roadway walls, the change in Excess pressure at the closed end becomes more complex. The measurements were taken in the explosion area. Since measurement point A is near the explosion source location, and gas explosions are instantaneous, measurement point A reaches its peak value instantaneously. As the explosion energy decays, the overall Excess pressure values at the measurement points exhibit nonlinear decay, ultimately tending towards a stable value. During a gas explosion, other measurement points are further away from the explosion source, and the Excess pressure measurement points reach their peak values sequentially. Due to continuous interference from the gas explosion shock waves, the shock waves are reflected and superimposed repeatedly at the closed end,presenting irregular shapes.

3.1.2 Analysis of Excess pressure at Axial Wall Measurement Points. As depicted in Figure 5, the axial data collection points in the tunnel are selected every 2 meters, with a total of four measurement points J, K, L, and M.

From Figure 6, it can be observed that after a gas explosion, shockwaves also exhibit reflection and superposition on the axial wall of the tunnel, leading to overall oscillations in the Excess pressure peak. The J and K measurement points display a decaying trend, while the L and M Excess pressure measurement points initially increase and then follow a decaying trend. This is due to the gas explosion not immediately reaching the air zone, resulting

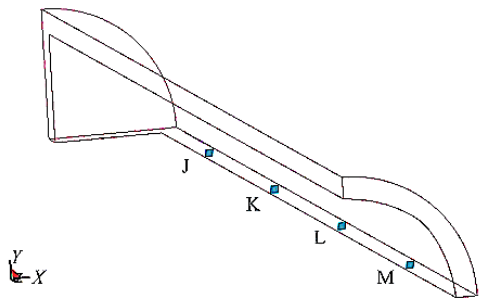

Figure 5: Axial data acquisition point.

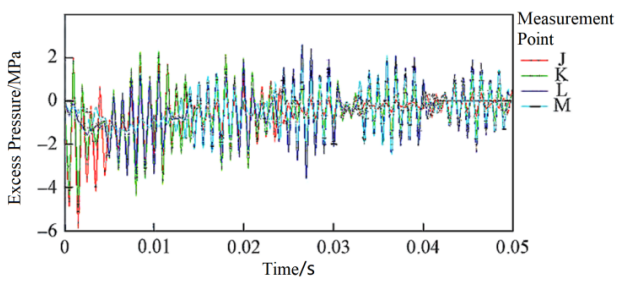

Figure 6: Axial wall excess pressure curve.

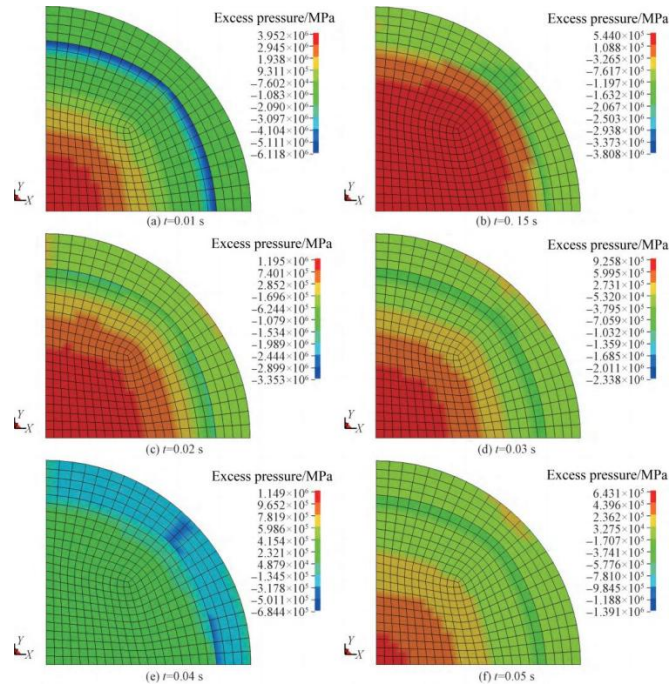

Figure 7: Closed-end wall overpressure change nephogram.

in fluctuations in the Excess pressure peak. Eventually, the Excess pressure from the gas explosion stabilizes.

It can be inferred that after a gas explosion, the shockwaves propagate outward, constrained by the axial walls. They travel along the inner walls, leading to an oscillating trend in axial Excess pressure. However, as the energy of the gas explosion decreases, the overall explosion Excess pressure decays and stabilizes. The maximum peak at measurement point L reaches 5.9 MPa, and the peak at measurement point K reaches 5.0 MPa, both exceeding those at measurement points L and M. This is because measurement points L and M are located in the air zone, where the shockwaves from the gas explosion do not directly reach the tunnel wall. These measurement points are also constrained by the tunnel wall, causing fluctuations in the Excess pressure measurement. Subsequently, the Excess pressure overall exhibits a decaying trend.

3.2 Wall Cloud Image Analysis

3.2.1 Analysis of Cloud Images at the Closed End. Figure 7 illustrates the cloud images of the overpressure variation on the wall surface at the closed end at different times. It can be observed that after the gas explosion, the locations with higher pressure at the closed end are the central and edge positions. The overpressure areas appear as rings and decrease gradually towards the perimeter. Subsequently, the overpressure areas continue to expand, covering the entire cross-section of the closed end. Due to the restriction of the wall surface at the closed end, the superposed overpressure reflection reaches a peak of 6.12 MPa, while the overpressure in the middle position of the wall surface is 3.95 MPa. Then, due to the action of negative pressure reduction at the closed end, the wall surface will experience secondary impacts, causing further damage to the wall surface.

In the cloud images at the closed end, the trend of stress variation over time can be observed. After the gas explosion, the larger explosion stress first appears in the central region of the closed end, where the surrounding degrees of freedom are restricted. The stress in the arched area is not significant, indicating that it experiences the least stress. As the reaction progresses, the stress on the wall surface at the closed end gradually accumulates, causing it to undergo damage and destruction.

3.2.2 Analysis of Axial Wall Cloud Diagram. Figure 8 depicts the cloud diagram of the overpressure change on the axial wall surface of the roadway. It can be observed from the figure that after the gas explosion, the shock wave load is initially applied to the inner wall surface of the roadway, and the overpressure effect prompts the wall structure to generate stress, gradually transmitting it to the outer wall. As the explosion reaction progresses, the energy gradually decays. Axially, the shock wave after the gas explosion propagates axially towards the opening direction. The shock wave is initially loaded on the wall structure of the gas area and gradually loaded on the wall surface of the air area. The wall surface of the gas area experiences the greatest load, which subsequently diffuses towards the wall surface of the air area.

The stress cloud diagram of the axial wall surface exhibits a ring-shaped distribution, spreading along the axial direction of the roadway. The stress loading is not uniformly distributed, and the stress variation on the wall surface increases over time, gradually developing from the edge region to the middle region of the wall

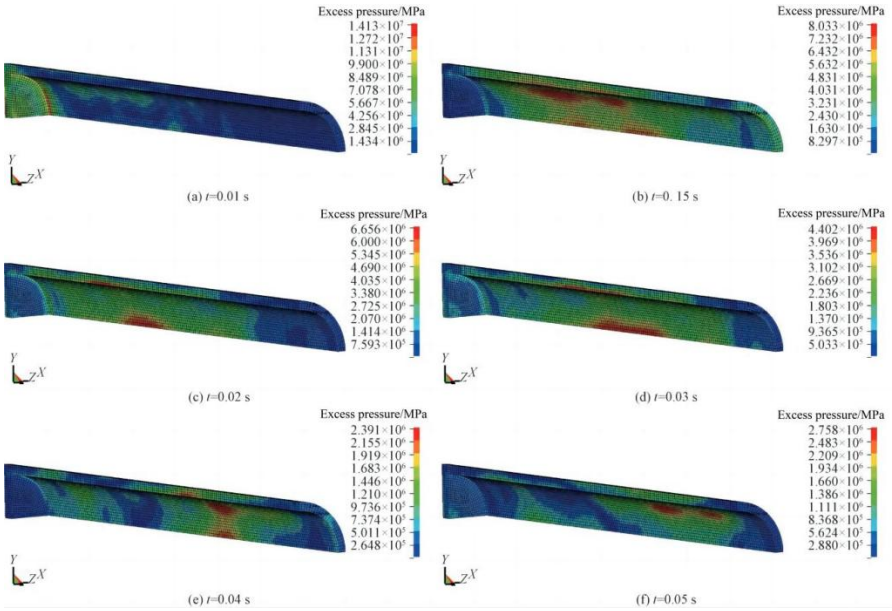

Figure 8: Axial wall change nephogram.

surface. Subsequently, a ring-shaped stress variation appears on the axial wall surface, with the explosion load continuously loading on the wall surface, but the stress continuously decreases. Stress concentration occurs in some regions, which tend to be the areas where damage and destruction first appear on the wall surface of the roadway.

3.2.3 Analysis of Wall Failure Characteristics. After a gas explosion, the shock wave initially impacts the closed end face, causing damage and destruction, with the most severe damage occurring in the middle position of the wall. This indicates that under the explosive load after the gas explosion, the tunnel wall structure undergoes damage and destruction that spreads axially along the wall structure as the reaction progresses. After the shock wave reaches the wall structure, the compression wave first propagates from the inner surface to the outer surface, and then the compression wave is stretched from the outer surface to form a tensile wave, which expands outwards along the tunnel wall structure.

Wall damage and failure are primarily the combined destructive results of dynamic failure by the explosion shock wave and quasi-static failure by detonation gases. The dynamic stress of the shock wave at the instant of explosion prompts the wall to generate circular compressive stress. As the energy of the explosion shock wave decays, a portion of the energy is lost through convection and heat exchange, while another portion is converted into stress waves. The wall structure undergoes tensile failure again under the superimposed effect of stress waves, and the stress wave loading intensifies, causing more severe damage. As the shock wave energy decreases, stress continues to be loaded onto the wall structure, resulting in static failure due to concentrated pressure on the wall structure, ultimately leading to tunnel instability and failure.

4 Conclusion

Through the use of ANSYS/LS-DYNA dynamics software to numerically simulate the gas explosion failure process in a tunnel, the process of Excess pressure and contour map changes during the explosion is visually reflected, leading to the following conclusions:

(1) The measured values of Excess pressure at the edge and center positions of the tunnel wall are relatively large, and the wall damage is relatively more severe. Therefore, the stacking of large objects should be avoided, and protection of the corresponding positions should be strengthened.

(2) After a gas explosion, shock waves may also exhibit reflection and superposition on the axial walls of the tunnel, leading to fluctuations in the overall peak Excess pressure.

(3) After a gas explosion, the shock wave propagates towards the opening, with the gas zone wall experiencing the maximum loading and gradually spreading towards the air zone.

(4) As the energy of the explosion shock wave attenuates, continuous stress loading on the wall structure leads to concentrated pressure exerting static damage on the wall structure, ultimately exceeding its capacity, resulting in instability and failure of the tunnel.

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